

Laboratorio de resistores en serie y en paralelo

Laboratorio de resistores en serie

$$m \times K = /$$

$$V \times K = m$$

$$V \times A = W$$

1) 2 resistores en serie con la siguiente información:

$$E = 9V$$

$$R_1 = 2k\Omega$$

$$R_2 = 900\Omega$$

$$I_T = \frac{9V}{2900\Omega} = 3.10 \text{ mA}$$

$$R_T = 2000 + 900 = 2900\Omega = 2.9K\Omega$$

$$V_T = (3.10 \text{ mA})(2.9K\Omega) = 8.99 \text{ V}$$

$$P_T = (3.10)(8.99) = 27.869 \text{ W}$$

2) 3 resistores en serie con la siguiente información:

$$E = 1.5V$$

$$R_1 = 500\Omega$$

$$R_2 = 700\Omega$$

$$I_T = \frac{E}{R_T} = \frac{1.5V}{0.6K\Omega} = 2.5 \text{ mA}$$

$$R_T = R_1 + R_2 = 500\Omega + 700\Omega = 600\Omega = 0.6K\Omega$$

$$V_T = I_T \times R_T = (2.5 \text{ mA})(0.6K\Omega) = 1.5 \text{ V}$$

$$P_T = I_T \times V_T = (2.5 \text{ mA})(1.5 \text{ V}) = 3.75 \text{ mW}$$

3) 2 resistores en serie con la siguiente información:

$$E = 3V$$

$$R_1 = 5K\Omega$$

$$R_2 = 10K\Omega$$

$$R_T = 5000 + 10000 = 15K\Omega$$

$$I_T = \frac{3V}{15K\Omega} = 0.2 \text{ mA}$$

$$V_T = (0.2 \text{ mA})(15K\Omega) = 3 \text{ V}$$

$$P_T = (0.2 \text{ mA})(3 \text{ V}) = 0.6 \text{ mW}$$

4) 3 resistores en serie con la siguiente información

$$E = 9V$$

$$R_1 = 5K\Omega$$

$$R_2 = 70K\Omega$$

$$R_3 = 75K\Omega$$

$$R_T = 5 + 70 + 75 = 30K\Omega$$

$$I_T = \frac{9V}{30K\Omega} = 0.3mA$$

$$V_T = (0.3mA)(30K\Omega) = 9V$$

$$P_T = (0.3mA)(9V) = 2.7mW$$

5) 3 resistores en serie con la siguiente información

$$E = 9V$$

$$R_1 = 50K\Omega$$

$$R_2 = 100K\Omega$$

$$R_3 = 1M\Omega$$

$$R_3 = \frac{1000000}{1000}$$

$$R_3 = 1000K\Omega$$

$$R_T = 50 + 100 + 1000 = 1150K\Omega$$

$$I_T = \frac{9V}{1150K\Omega} = 7.8260 \times 10^{-3} mA$$

$$V_T = (7.8260 \times 10^{-3})(1150K\Omega) = 9V$$

$$P_T = (7.8260 \times 10^{-3})(9V) = 0.070434mW$$

6) 4 resistores en serie con la siguiente información

$$E = 9V$$

$$R_1 = 2K\Omega$$

$$R_2 = 2K\Omega$$

$$R_3 = 72K\Omega$$

$$R_4 = 100K\Omega$$

$$R_T = 2 + 2 + 72 + 100 = 176K\Omega$$

$$I_T = \frac{9V}{176K\Omega} = 0.07758mA$$

$$V_T = (0.07758mA)(176K\Omega) = 5.99928 \approx 9V$$

$$P_T = (0.07758)(9V) = 0.6982mW$$

7) Resistores en serie con la siguiente información

$$E = 9V$$

$$R_1 = 80\Omega$$

$$R_2 = 20\Omega$$

$$R_3 = 40K\Omega$$

$$R_4 = 10\Omega$$

$$R_T = 0.08 + 0.02 + 0.04 + 0.1 = 0.24K\Omega$$

$$I_T = \frac{9V}{0.24K\Omega} = 37.5mA$$

$$V_T = (37.5mA)(0.24K\Omega) = 9V$$

$$P_T = (37.5mA)(9V) = 337.5mW$$

Resistores en paralelo

$$1) R_T = \frac{R_1 \# R_2}{R_1 + R_2} = \frac{2K\Omega \# 900\Omega}{2K\Omega + 900\Omega} = 0.62K\Omega$$

$$I_T = \frac{E}{R_T} = \frac{9V}{0.62K\Omega} = 14.52mA$$

$$E = V_1 = V_2 = 9V$$

$$I_1 = \frac{V_1}{R_1} = \frac{9V}{2K\Omega} = 4.5mA$$

$$I_2 = \frac{V_2}{R_2} = \frac{9V}{0.9K\Omega} = 10mA$$

$$P_1 = V_{R1} \# I_{R1} = 9 \# 4.5 = 40.5mW$$

$$P_2 = V_{R2} \# I_{R2} = 9 \# 10 = 90mW$$

$$2) R_T = \frac{R_1 \# R_2}{R_1 + R_2} = \frac{500 \# 300}{500 + 300} = 83.3 \Omega$$

$$I_T = \frac{E}{R_T} = \frac{7.5V}{83.3} = 0.078A$$

$$E = V_1 = V_2 = 7.5V$$

$$I_1 = \frac{V_1}{R_1} = \frac{7.5V}{300\Omega} = 0.003A$$

$$I_2 = \frac{V_2}{R_2} = \frac{7.5V}{100\Omega} = 0.075A$$

$$P_1 = V_{R_1} \# I_1 = 7.5V \# 0.003A = 0.0045W$$

$$P_2 = V_{R_2} \# I_2 = 7.5V \# 0.075A = 0.0225W$$

$$3) R_T = \frac{R_1 \# R_2}{R_1 + R_2} = \frac{5K\Omega \# 10K\Omega}{5 + 10} = 3.33K\Omega$$

$$I_T = \frac{E}{R_T} = \frac{3V}{3.33K\Omega} = 0.90mA$$

$$I_1 = \frac{V_{R_1}}{R_1} = \frac{3V}{5K\Omega} = 0.6mA$$

$$E = V_{R_1} = V_{R_2} = 3V$$

$$I_2 = \frac{V_{R_2}}{R_2} = \frac{3V}{10K\Omega} = 0.3mA$$

$$P_1 = V_{R_1} \# I_1 = 3V \# 0.6mA = 1.8mW$$

$$P_2 = V_{R_2} \# I_2 = 3V \# 0.3mA = 0.9mW$$

$$4) R_T = \frac{R_1 \# R_2 \# R_3}{R_1 \# R_2 + R_3 + R_1 \# R_3 + R_2 \# R_3} = \frac{5K\Omega \# 10K\Omega \# 15K\Omega}{5 \# 10 + 5 \# 15 + 10 \# 15} = 2.72K\Omega$$

$$I_T = \frac{E}{R_T} = \frac{9V}{2.72K\Omega} = 3.31mA$$

$$I_1 = \frac{V_{R_1}}{R_1} = \frac{9V}{5K\Omega} = 1.8mA$$

$$E = V_1 = V_2 = V_3 = 9V$$

$$I_2 = \frac{V_{R_2}}{R_2} = \frac{9V}{10K\Omega} = 0.9mA$$

$$I_3 = \frac{V_{R_3}}{R_3} = \frac{9V}{15K\Omega} = 0.6mA$$

$$P_1 = V_{R_1} \# I_1 = 9V \# 1.8mA = 16.2mW$$

$$P_2 = V_{R_2} \# I_2 = 9V \# 0.9mA = 8.1mW$$

$$P_3 = V_{R_3} \# I_3 = 9V \# 0.6mA = 5.4mW$$

$$5) R_T = \frac{R_1 \parallel R_2 \parallel R_3}{R_1 \parallel R_2 \parallel R_3} = \frac{50k\Omega \parallel 100k\Omega \parallel 7M}{50k\Omega \parallel 100k\Omega \parallel 7M} = 32.25k\Omega$$

$$I_T = \frac{E}{R_T} = \frac{9V}{32.25k\Omega} = 0.28mA$$

$$I_1 = \frac{V_1}{R_1} = \frac{9V}{50k\Omega} = 0.18mA$$

$$I_2 = \frac{V_2}{R_2} = \frac{9V}{100k\Omega} = 0.09mA$$

$$I_3 = \frac{V_3}{R_3} = \frac{9V}{7M} = 0.009mA$$

$$E = V_1 = V_2 = V_3 = 9V$$

$$P_1 = V_1 \cdot I_1 = 9V \cdot 0.18mA = 1.62mW$$

$$P_2 = V_2 \cdot I_2 = 9V \cdot 0.09mA = 0.81mW$$

$$P_3 = V_3 \cdot I_3 = 9V \cdot 0.009mA = 0.081mW$$

$$6) R_T = \frac{R_1 \parallel R_2}{R_1 + R_2} + \frac{R_3 \parallel R_4}{R_3 + R_4} = \frac{2k\Omega \parallel 2k\Omega}{2 + 2} + \frac{12k\Omega \parallel 100k\Omega}{12 + 100} = 0.91k\Omega$$

$$I_T = \frac{E}{R_T} = \frac{9V}{0.91k\Omega} = 9.89mA$$

$$E = V_1 = V_2 = V_3 = V_4 = 9V$$

$$I_1 = \frac{9V}{2k\Omega} = 4.5mA$$

$$I_2 = \frac{9V}{12k\Omega} = 0.75mA$$

$$I_3 = \frac{9V}{2k\Omega} = 4.5mA$$

$$I_4 = \frac{9V}{100k\Omega} = 0.09mA$$

$$P_1 = 9V \cdot 4.5mA = 40.5mW$$

$$P_2 = 9V \cdot 0.75mA = 6.75mW$$

$$P_3 = 9V \cdot 4.5mA = 40.5mW$$

$$P_4 = 9V \cdot 0.09mA = 0.81mW$$

$$7) R_T = \frac{R_1 \parallel R_2}{R_1 + R_2} + \frac{R_3 \parallel R_4}{R_3 + R_4} = \frac{80\Omega \parallel 20\Omega}{80 + 20} + \frac{40k\Omega \parallel 100\Omega}{40 + 100} = 13.79\Omega$$

$$I_T = \frac{E}{R_T} = \frac{9V}{13.79\Omega} = 0.65A$$

$$E = V_1 = V_2 = V_3 = V_4 = 9V$$

$$I_1 = 9V / 80\Omega = 0.11A$$

$$I_3 = 9V / 40k\Omega = 0.225mA$$

$$I_2 = 9V / 20\Omega = 0.45A$$

$$I_4 = 9V / 100\Omega = 0.09A$$

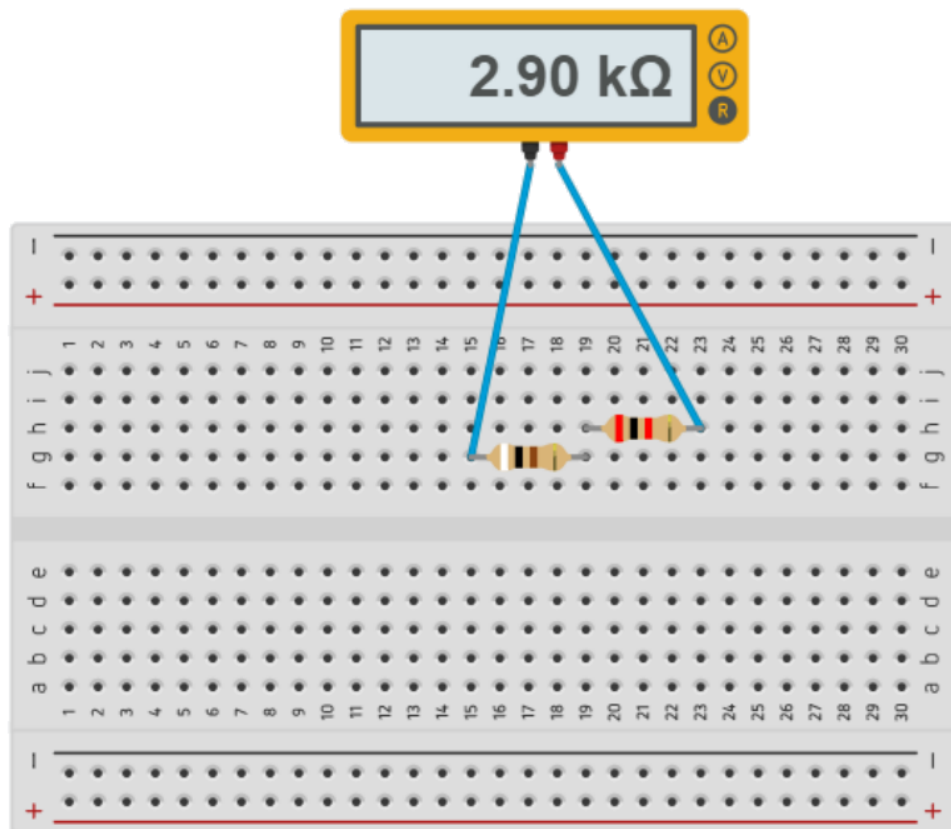
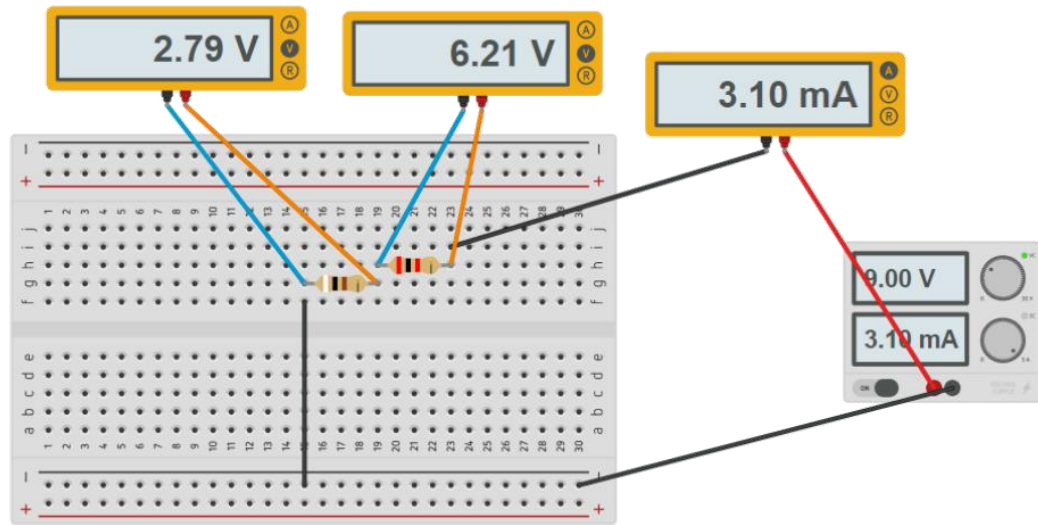
$$P_1 = 9V \cdot 0.11A = 0.99W$$

$$P_3 = 9V \cdot 0.225mA = 2.025mW$$

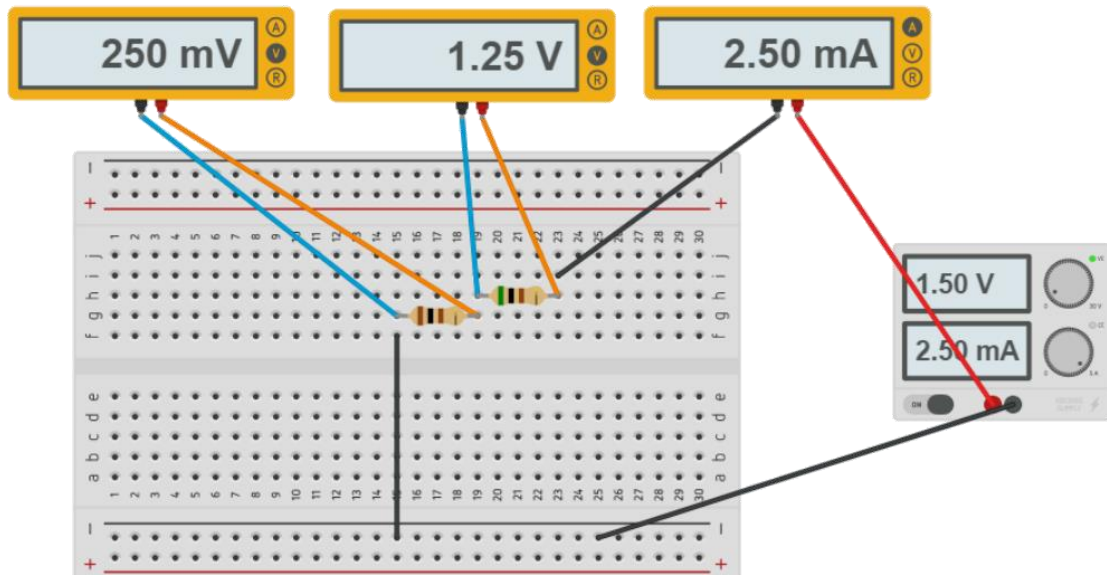
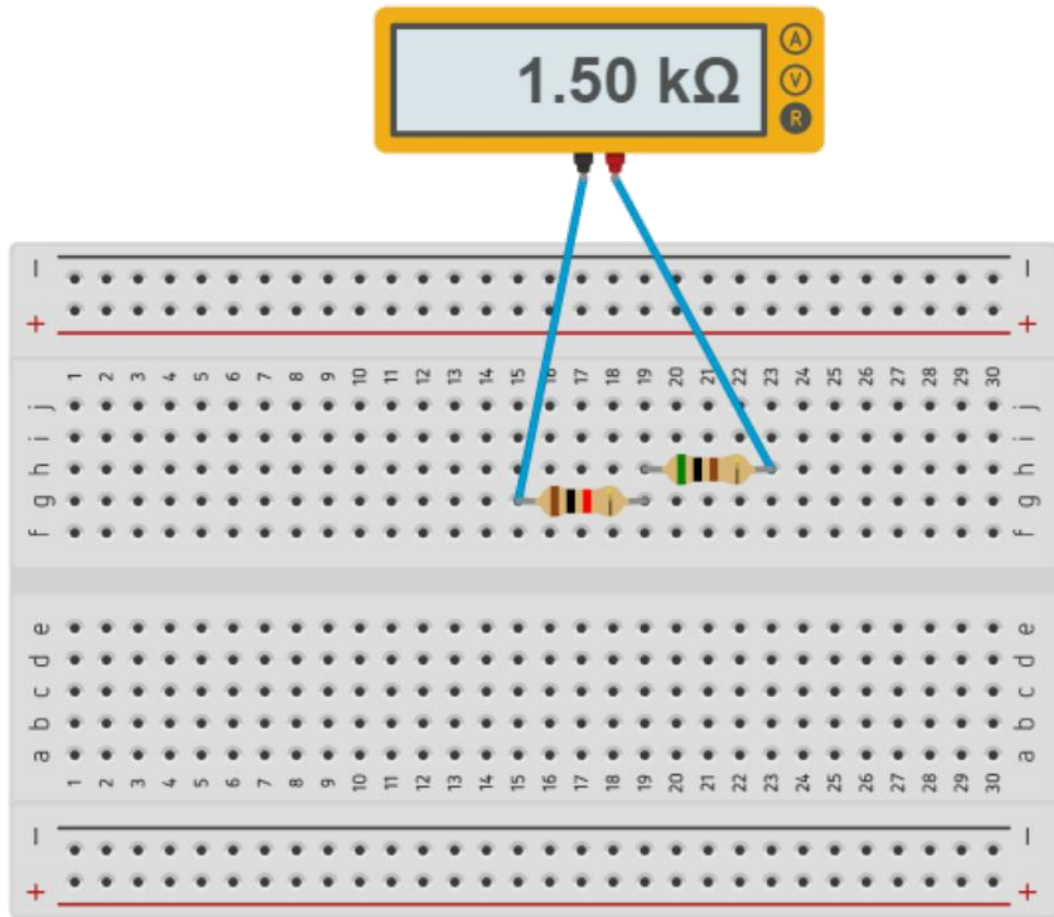
$$P_2 = 9V \cdot 0.45A = 4.05W$$

$$P_4 = 9V \cdot 0.09A = 0.81W$$

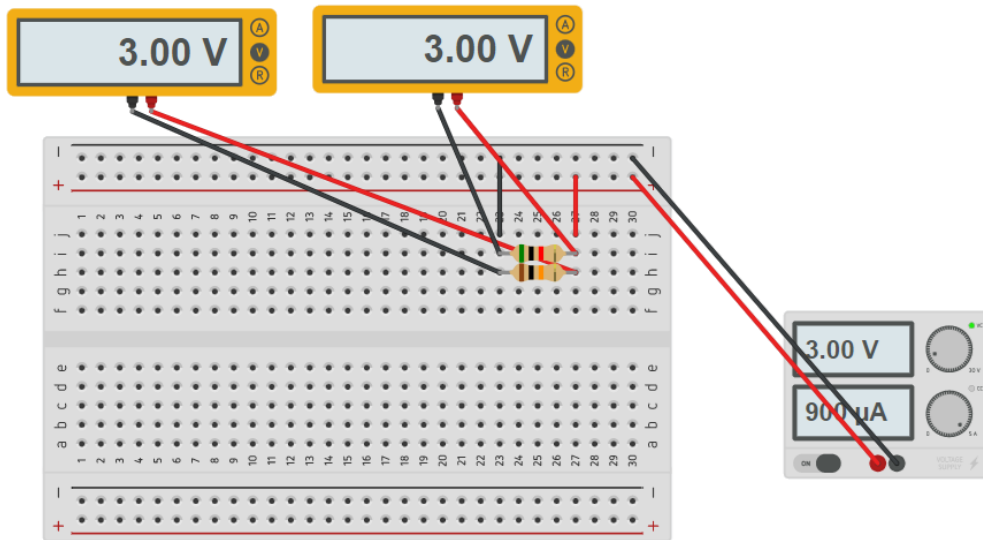
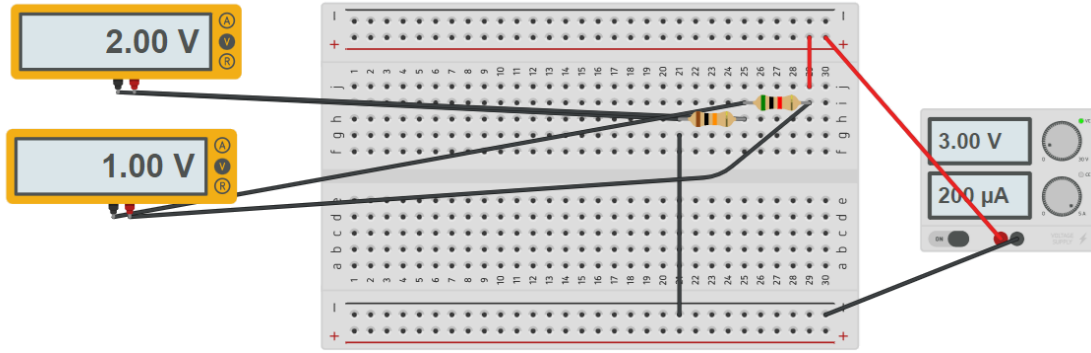
Ejercicio 1



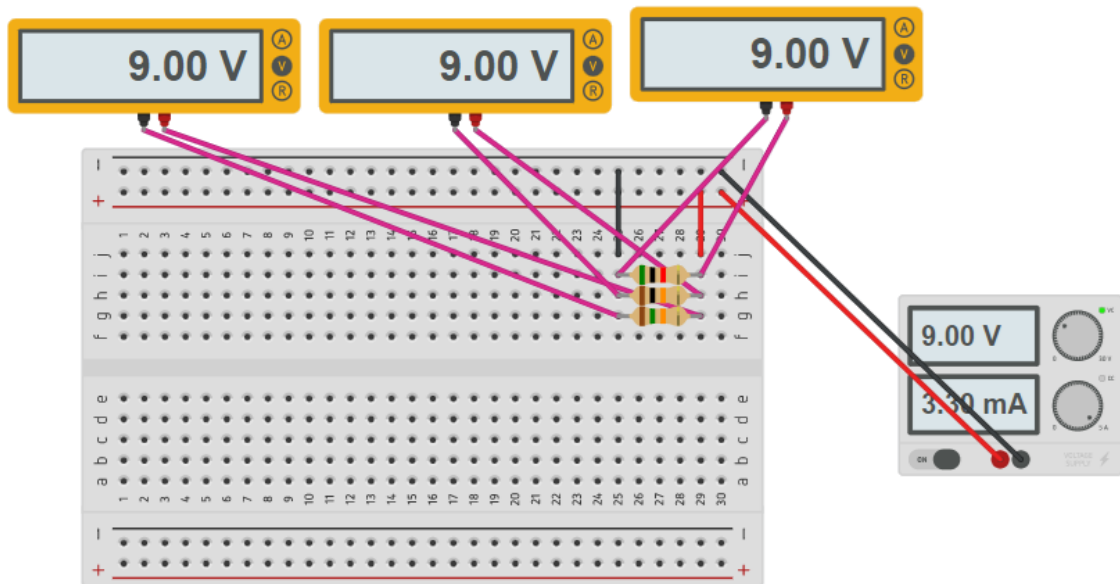
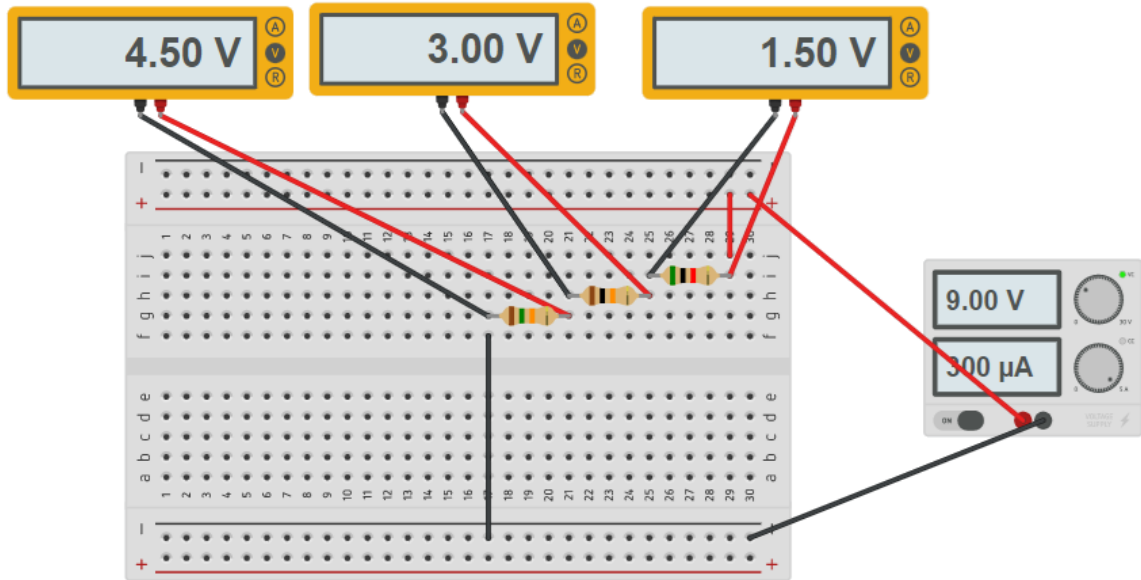
Ejercicio 2



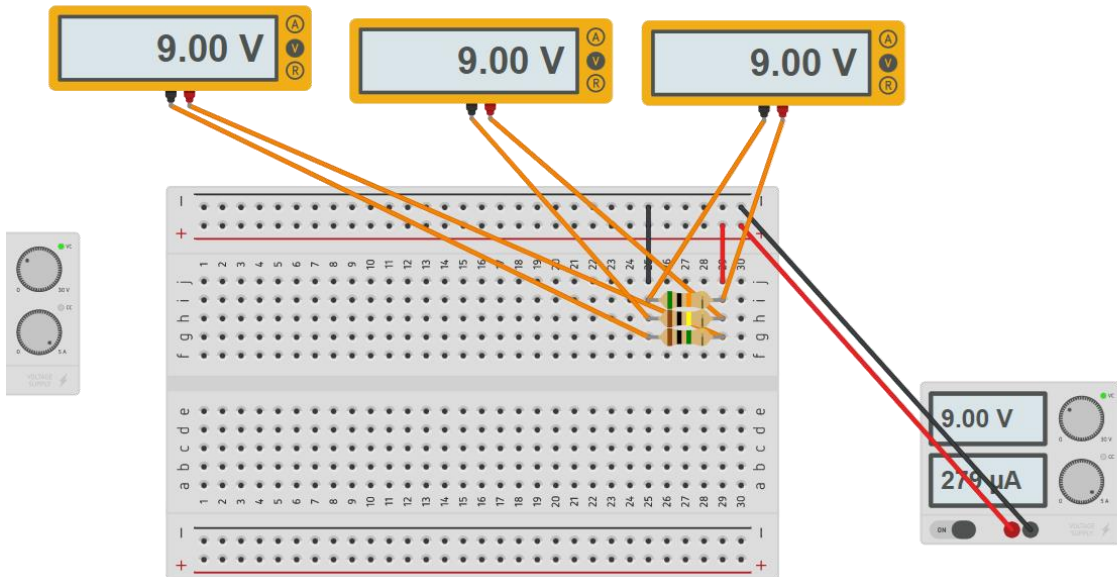
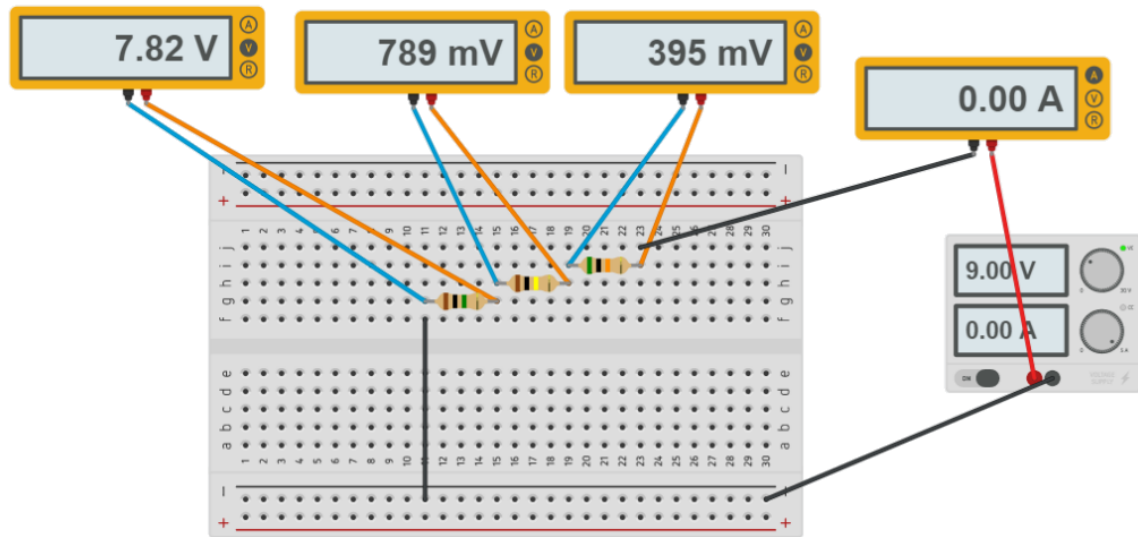
Ejercicio 3



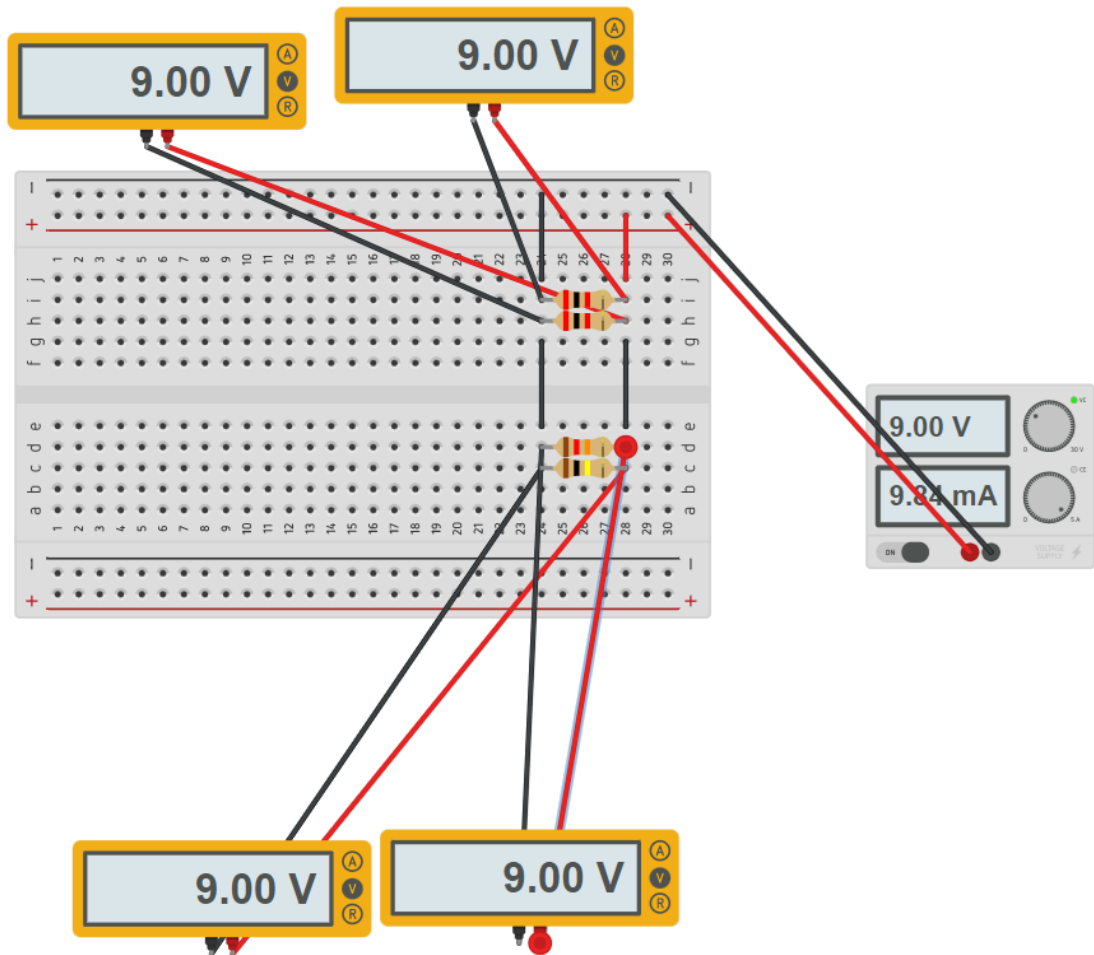
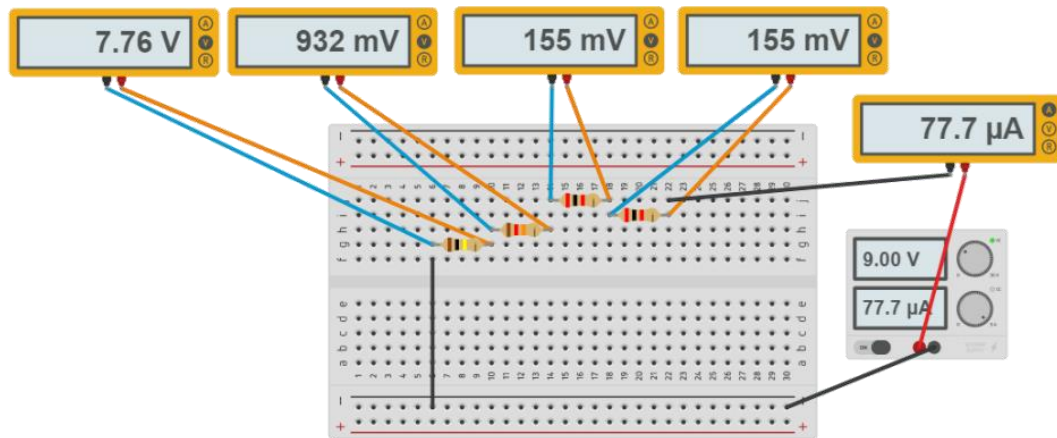
Ejercicio 4



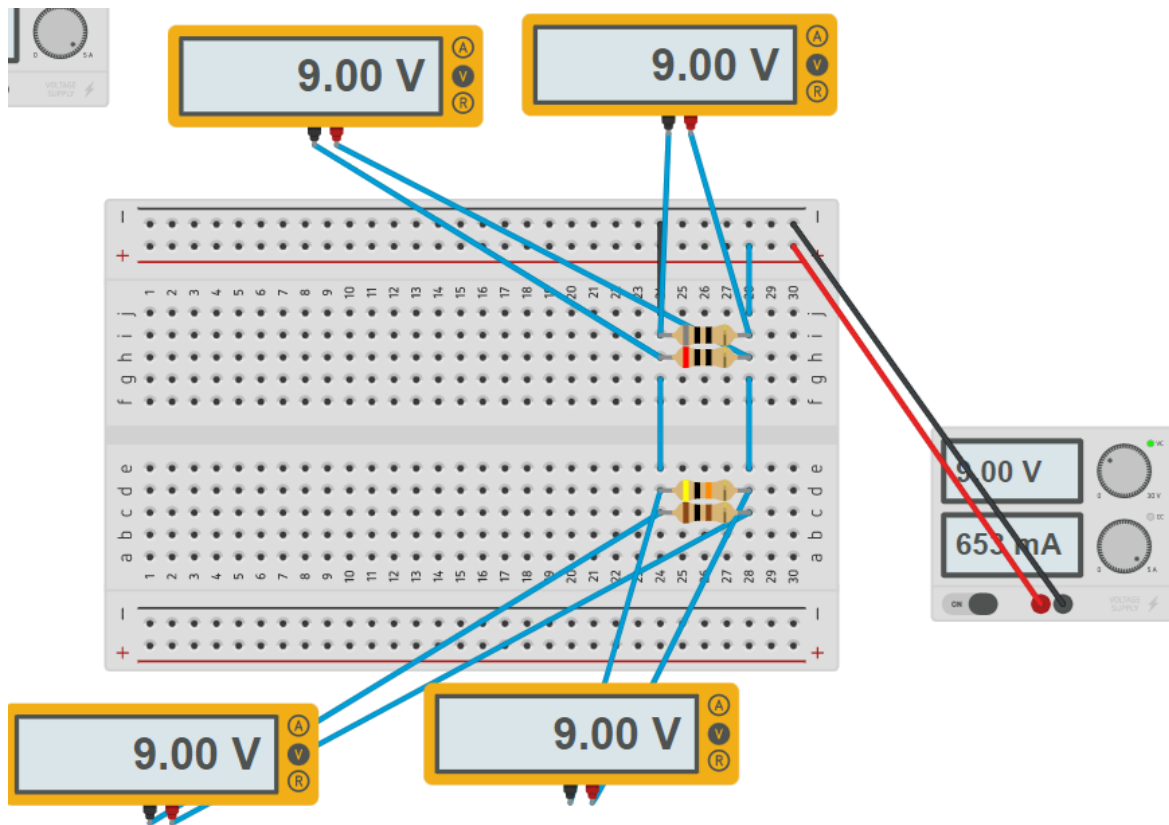
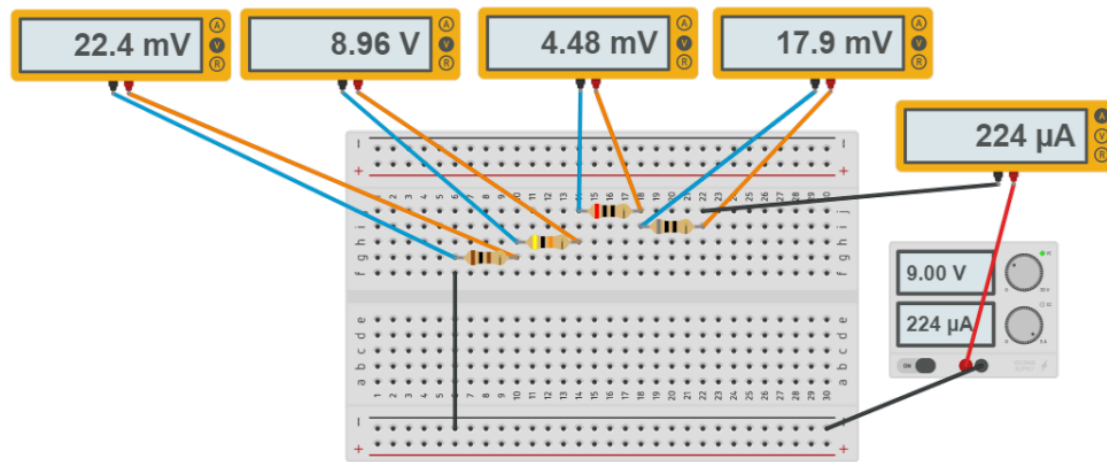
Ejercicio 5



Ejercicio 6



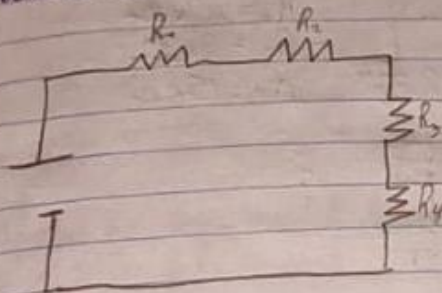
Ejercicio 7



Laboratorio Circuitos Mixtos

Laboratorio

Ejercicio 1



$$E = V = 20 \text{ Volts}$$

$$R_1 = 100 \text{ k}\Omega$$

$$R_2 = 50 \text{ k}\Omega$$

$$R_3 = 75 \text{ k}\Omega$$

$$R_4 = 100 \text{ k}\Omega$$

Paso 1

$$R_T = R_1 + R_2 + R_3 + R_4 = 100 \text{ k}\Omega + 50 \text{ k}\Omega + 75 \text{ k}\Omega + 100 \text{ k}\Omega = 325 \text{ k}\Omega$$

Paso 2

$$I = \frac{E}{R_T} = \frac{20 \text{ V}}{325 \text{ k}\Omega} = 0.062 \text{ mA}$$

Paso 3

$$V_1 = I \cdot R_1 = 0.062 \text{ mA} \cdot 100 \text{ k}\Omega = 6.2 \text{ V}$$

$$V_2 = 0.062 \cdot 50 = 3.1 \text{ V}$$

$$V_3 = 0.062 \cdot 75 = 4.65 \text{ V}$$

$$V_4 = 0.062 \cdot 100 = 6.2 \text{ V}$$

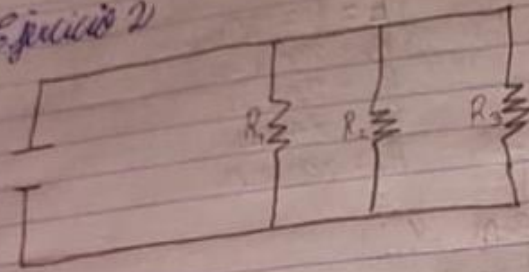
$$P_1 = I \cdot V_1 = 0.062 \text{ mA} \cdot 6.2 \text{ V} = 0.3844 \text{ mW}$$

$$P_2 = 0.062 \text{ mA} \cdot 3.1 \text{ V} = 0.1922 \text{ mW}$$

$$P_3 = 0.062 \text{ mA} \cdot 4.65 \text{ V} = 0.2883 \text{ mW}$$

$$P_4 = 0.062 \text{ mA} \cdot 6.2 \text{ V} = 0.3844 \text{ mW}$$

Ejercicio 2



$$\begin{aligned} E &= 50V \\ R_1 &= 10k\Omega \\ R_2 &= 2k\Omega \\ R_3 &= 15k\Omega \end{aligned}$$

Paso 1

$$R_T = \frac{R_1 \cdot R_2 \cdot R_3}{R_1 \cdot R_2 + R_1 \cdot R_3 + R_2 \cdot R_3} = \frac{10k\Omega \cdot 2k\Omega \cdot 15k\Omega}{10 \cdot 2 + 10 \cdot 15 + 2 \cdot 15} = 1.5k\Omega$$

Paso 2

$$I = \frac{E}{R_T} = \frac{50V}{2.5k\Omega} = 33.33mA$$

Paso 3

$$V_1 = V_2 = V_3 = 50V$$

$$I_1 = \frac{V_1}{R_1} = \frac{50V}{10k\Omega} = 5mA$$

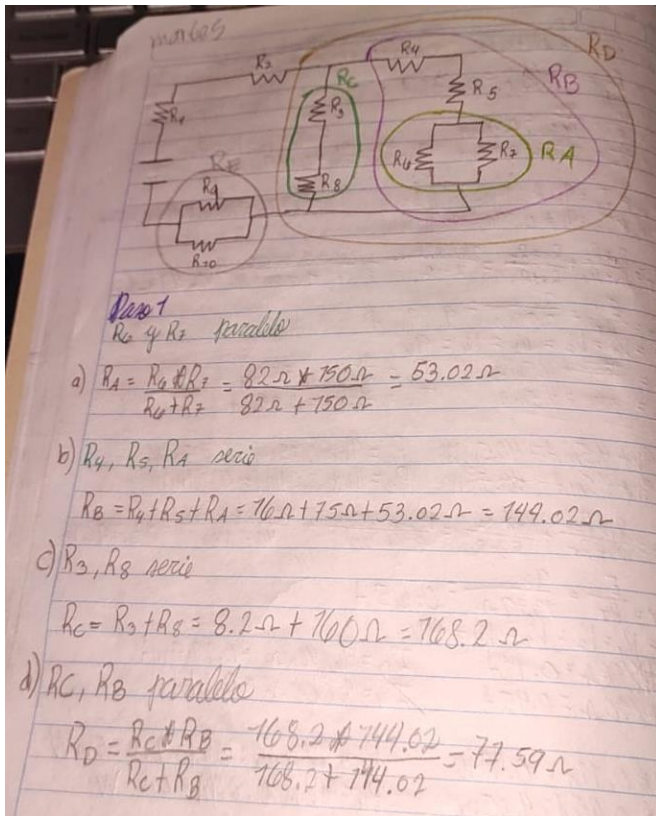
$$I_2 = \frac{V_2}{R_2} = \frac{50V}{2k\Omega} = 25mA$$

$$I_3 = \frac{V_3}{R_3} = \frac{50V}{15k\Omega} = 3.33mA$$

$$P_1 = V \cdot I_1 = 50V \cdot 5mA = 250mW$$

$$P_2 = V \cdot I_2 = 50V \cdot 25mA = 1250mW$$

$$P_3 = V \cdot I_3 = 50V \cdot 3.33mA = 166.5mW$$



e) R_1, R_{10} paralelo

$R_E = \frac{R_1 \parallel R_{10}}{R_1 + R_{10}} = \frac{51 \Omega \cdot 120 \Omega}{51 \Omega + 120 \Omega} = 36.63 \Omega$

f) R_1, R_2, R_D, R_E série

$R_G = R_1 + R_2 + R_D + R_E = 7 + 3 + 77.59 \Omega + 36.63 \Omega = 4.77 \text{ k}\Omega$

Passo 2

$I = \frac{E}{R_G} = \frac{20V}{4.77 \text{ k}\Omega} = 4.87 \text{ mA}$

Exercício 3

$V_{R10} = V_{R8} = V_{R6} = 48V$

$I_{R8} = \frac{V_{R8}}{R_{R8}} = \frac{48V}{6 \Omega} = 8A$

$I_{R6} = \frac{V_{R6}}{R_{R6}} = \frac{48V}{12 \Omega} = 4A$

$R_C = V_{R10} = I_{R10} \cdot R_{10} = 4A \cdot 10 \Omega = 40V$

$V_2 = I_{R2} \cdot R_2 = 4A \cdot 2 \Omega = 8V$

$R_B = V_4 = I_B \cdot R_4 = 8A \cdot 4 \Omega = 32V$

$V_2 = I_B \cdot R_2 = 8A \cdot 2 \Omega = 16V$

Potências

1) $P_1 = I \cdot V = 12A \cdot 60V = 720W$

2) $P_2 = 12A \cdot 24V = 288W$

3) $P_3 = 12A \cdot 36V = 432W$

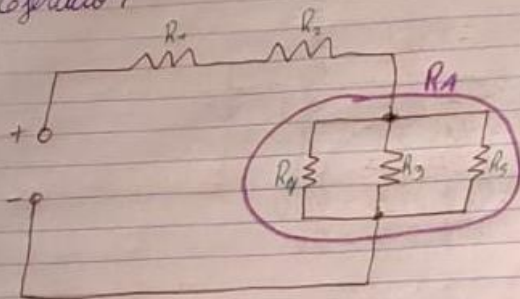
4) $P_4 = I_{R8} \cdot V_4 = 8A \cdot 32V = 256W$

5) $P_5 = 8A \cdot 16V = 128W$

6) $P_6 = I_{R6} \cdot V_{R6} = 4A \cdot 40V = 160W$

7) $P_7 = I_{R6} \cdot V_2 = 4A \cdot 8V = 32W$

Ejercicio 4



$$\begin{aligned} E &= 50V \\ R_1 &= 3\Omega \\ R_2 &= 7\Omega \\ R_3 &= 6\Omega \\ R_4 &= 4\Omega \\ R_5 &= 12\Omega \end{aligned}$$

Paso 1

$$R_A = \frac{R_4 \cdot R_3 \cdot R_5}{R_4 \cdot R_3 + R_4 \cdot R_5 + R_3 \cdot R_5} = \frac{4 \cdot 6 \cdot 12}{4 \cdot 6 + 4 \cdot 12 + 6 \cdot 12} = 2\Omega$$

$$R_T = R_1 + R_2 + R_A = 3\Omega + 7\Omega + 2\Omega = 12\Omega$$

Paso 2

$$I = \frac{E}{R_T} = \frac{50V}{12\Omega} = 4.17A$$

Paso 3

$$V_3 = I \cdot R_3 = 4.17A \cdot 3\Omega = 12.51V$$

$$V_7 = I \cdot R_7 = 4.17A \cdot 7\Omega = 29.19V$$

$$V_{R_A} = I \cdot R_A = 4.17A \cdot 2\Omega = 8.34V$$

$$V_{R_4} = I_A = \frac{V_{R_A}}{R_4} = \frac{8.34V}{4\Omega} = 2.08A$$

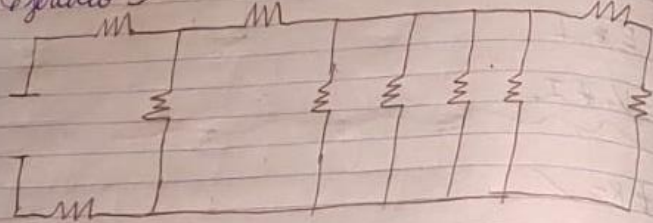
$$I_6 = \frac{V_{R_A}}{R_3} = \frac{8.34V}{6\Omega} = 1.39A$$

$$I_{12} = \frac{V_{R_A}}{R_{12}} = \frac{8.34V}{12\Omega} = 0.70A$$

Potencias

$$\begin{aligned} P_3 &= I \cdot V = 4.17A \cdot 12.51V = 52.17W \\ P_7 &= I \cdot V = 4.17A \cdot 29.19V = 121.72W \\ P_4 &= I \cdot V = 2.08A \cdot 8.34V = 17.35W \\ P_6 &= I \cdot V = 1.39A \cdot 8.34V = 11.59W \\ P_{12} &= I \cdot V = 0.70A \cdot 8.34V = 5.83W \end{aligned}$$

Ejercicio 5



Paso 1

$$R_a = 1 + 1 = 2 \text{ k}\Omega$$

$$R_b = \frac{10 \parallel 20 \parallel 2}{10 \parallel 20 + 10 \parallel 2 \parallel 20 \parallel 2} = 1.5 \text{ k}\Omega$$

$$R_c = \frac{1.5 \parallel 12 \parallel 10}{1.5 \parallel 12 + 1.5 \parallel 10 + 12 \parallel 10} = 1.2 \text{ k}\Omega$$

$$R_d = 2 + 1.2 = 3.2 \text{ k}\Omega$$

$$R_e = \frac{8 \parallel 3.2}{8 + 3.2} = 2.28 \text{ k}\Omega$$

$$R_t = 10 + 5 + 2.28 = 17.28$$

Paso 2

$$I = \frac{30}{17.28} = 1.74$$

Paso 3

$$V = I \cdot R$$

$$1.74 \cdot 10 = 17.4 \text{ V}_1$$

$$1.74 \cdot 5 = 8.7 \text{ V}_2$$

$$1.74 \cdot 2 = 3.48 \text{ V}_3$$

$$1.24 \cdot 2 = 2.42 \text{ V}_4$$

$$1.24 \cdot 1.2 = 1.45 \text{ V}_5$$

$$0.72 \cdot 1 = 0.72 \text{ V}_6$$

$$0.72 \cdot 1 = 0.72 \text{ V}_7$$

$$P = V \cdot I$$

$$P_1 = 17.4 \cdot 1.74 = 30.28$$

$$P_2 = 8.7 \cdot 1.74 = 15.14$$

$$P_3 = 3.48 \cdot 1.74 = 6.05$$

$$P_4 = 2.42 \cdot 1.74 = 4.21$$

$$P_5 = 1.45 \cdot 1.2 = 1.74$$

$$P_6 = 1.45 \cdot 0.74 = 1.07$$

$$P_7 = 1.45 \cdot 0.74 = 1.07$$

$$P_8 = 0.72 \cdot 1.74 = 1.25$$

$$P_{10} = 0.72 \cdot 1.74 = 1.25$$

$$I = V/R$$

$$3.97 / 8 = .5 \text{ R}_3$$

$$3.97 / 3.2 = 1.24 \text{ R}_4$$

$$1.45 / 12 = .12 \text{ R}_5$$

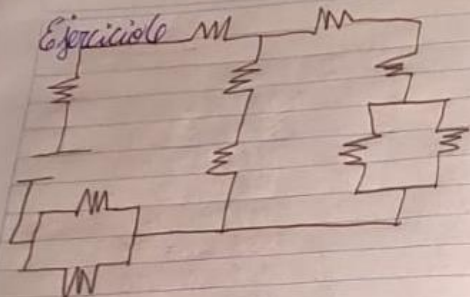
$$1.45 / 10 = .14 \text{ R}_6$$

$$1.45 / 10 = .14 \text{ R}_7$$

$$1.45 / 20 = .07 \text{ R}_8$$

$$1.45 / 2 = .72$$

Ejercicio 6



Paso 1

$$R_1 = \frac{82 \times 150}{82 + 150} = 53.02 \Omega$$

$$R_2 = 100 + 75 + 53.02 = 144.02$$

$$R_3 = 8.2 + 160 = 168.2$$

$$R_4 = \frac{168.2 \times 144.02}{168.2 + 144.02} = 77.54 \Omega$$

$$R_5 = \frac{51 + 130}{51 + 130} = 36.63 \Omega$$

$$R_6 = 7 + 3 + \frac{77.54}{1000} + \frac{36.63}{1000} = 4.77$$

Paso 2

$$I = \frac{20}{4.11} = 4.87 \text{ mA}$$

Paso 3

$$R_1 = V = I \times R = 4.87 \times 1 = 4.87 \text{ V}$$

$$R_2 = 4.87 \times 36.63 = 0.78 \text{ V}$$

$$R_3 = 4.87 \times 36.63 = 0.78 \text{ V}$$

$$R_4 = 4.87 \times 77.51 = 0.38 \text{ V}$$

$$I = 18/51 = 3.53$$

$$I = 16/130 = 1.38$$

$$I = 24/168.2 = 2.24$$

$$I = 38/144.02 = 2.63$$

$$0.14/8.2 = 7.7 \text{ mA}$$

$$0.70/150 = 9.33 \text{ mA}$$

$$220 \times 8.2 = 0.019 \text{ V}$$

$$6.24 \times 160 = 0.36 \text{ V}$$

$$2.63 \times 76 = 0.42 \text{ V}$$

$$2.63 \times 75 = 0.20 \text{ V}$$

$$2.63 \times 53.02 = 0.79 \text{ V}$$

Potencia

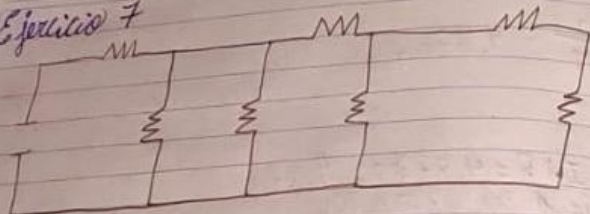
$$4.87 \times 4.87 = 23.72 \text{ W}$$

$$W = 4.87 \times 14.61 = 71.75$$

$$W = 4.87 \times 7.8 = 0.88 \text{ W}$$

$$W = 4.87 \times 0.38 = 1.95 \text{ W}$$

Ejercicio 7



Paso 1

$$R_A = 2 + 8 = 10 \Omega$$

$$R_B = \frac{10 \times 10}{10 + 10} = 5 \Omega$$

$$R_C = 5 + 1 = 6 \Omega$$

$$R_D = \frac{1}{\frac{1}{12} + \frac{1}{4} + \frac{1}{6}} = 2 \Omega$$

$$R_E = 1 + 2 = 3 \Omega = \frac{3}{1000} = 0.003 K \Omega$$

Paso 2

$$I = \frac{25}{0.003} = 8333.33$$

$$25 \times 8333.33 = 208185.25$$

Paso 3

$$R_1 = V = I \times R = 8333.33 \times 0.001 = 8.33$$

$$R_2 = 8333.33 \times 0.002 = 16.66$$

$$R_4 = 2776.67 \times 0.001 = 2.78$$

$$R_8 = 2776.67 \times 0.005 = 13.88$$

$$R_6 = 1388 \times 0.002 = 2.78$$

$$R_7 = 1388 \times 0.008 = 11.10$$

$$R_2 = I = V/R = 16.66 / 0.012 = 1388.33$$

$$R_3 = 16.66 / 0.004 = 4165$$

$$R_6 = 16.66 / 0.006 = 2776.67$$

$$R_4 = 13.88 / 0.01 = 1388$$

$$R_5 = 13.88 / 0.01 = 1388$$

$$R_1 = P = V \times I = 8.33 \times 8333.33 = 69416.64$$

$$R_2 = 16.66 \times 1388.33 = 23129.58$$

$$R_3 = 16.66 \times 4165 = 69388.9$$

$$R_4 = 2.78 \times 2776.67 = 7719.14$$

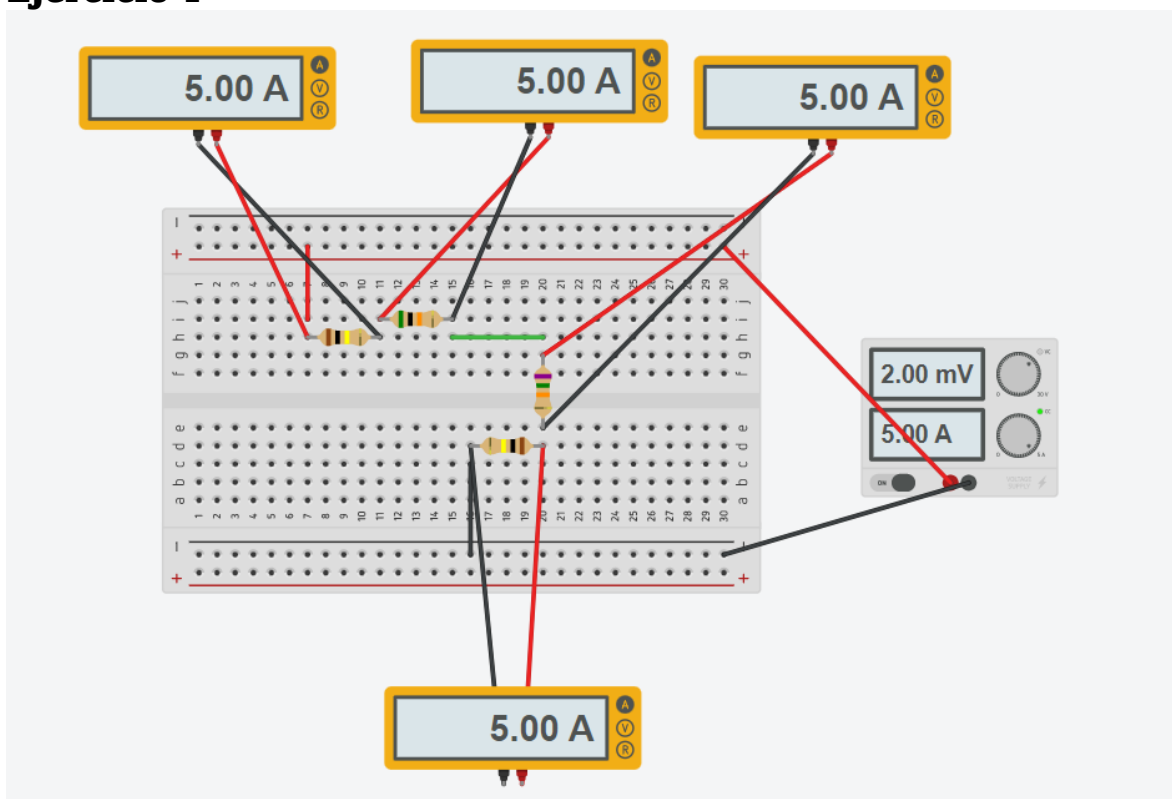
$$R_5 = 13.88 \times 1388 = 19265.44$$

$$R_6 = 2.78 \times 1388 = 3858.64$$

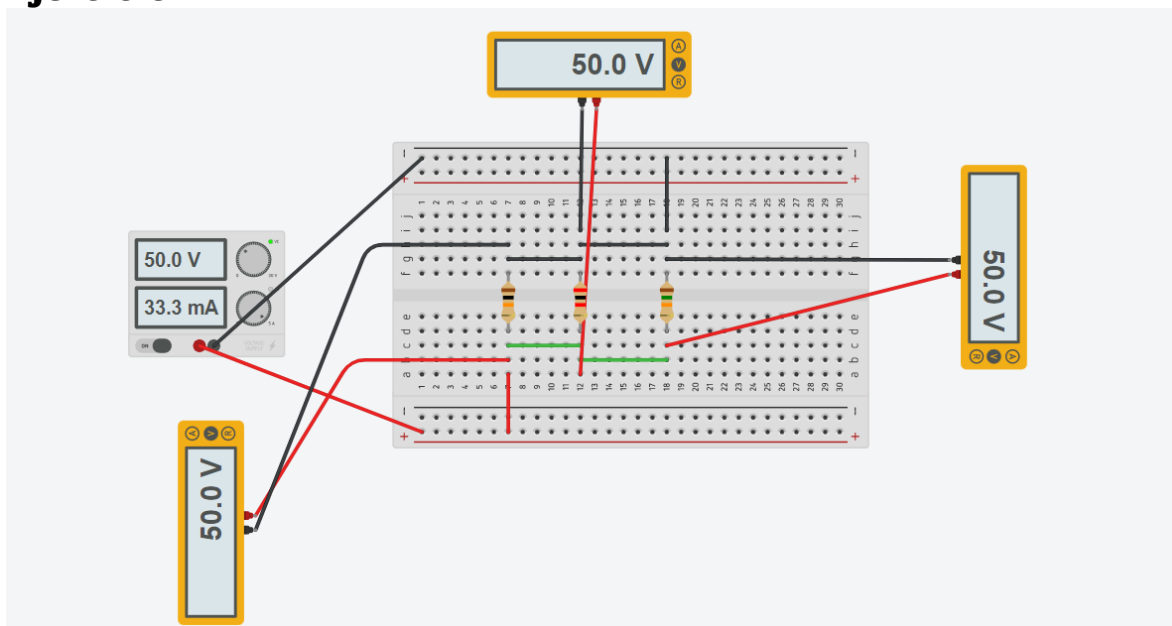
$$R_7 = 11.10 \times 1388 = 15406.8$$

$$208185.25$$

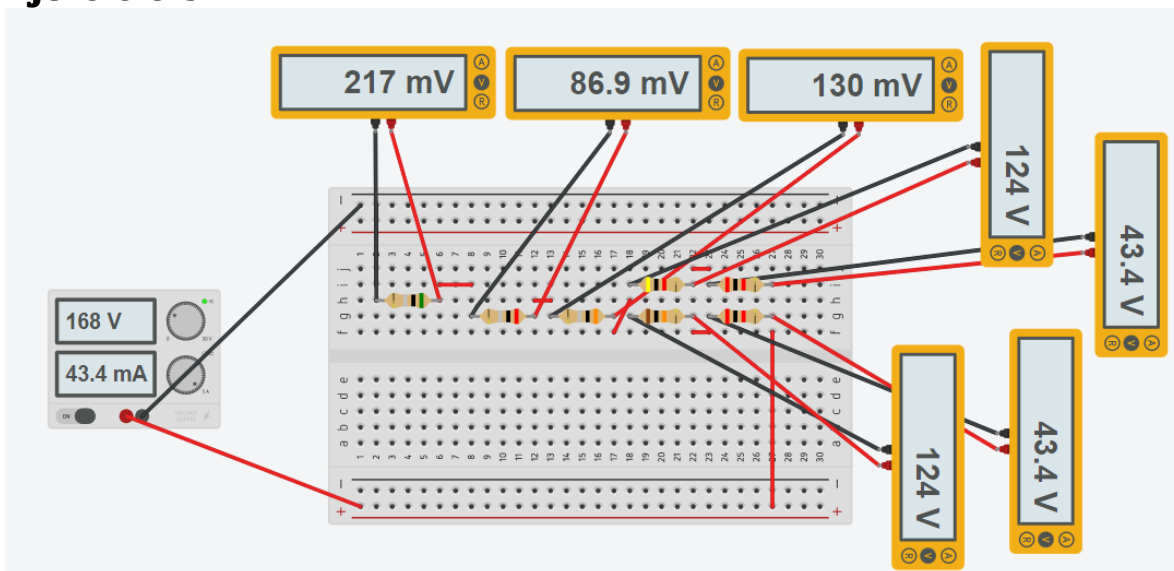
Ejercicio 1



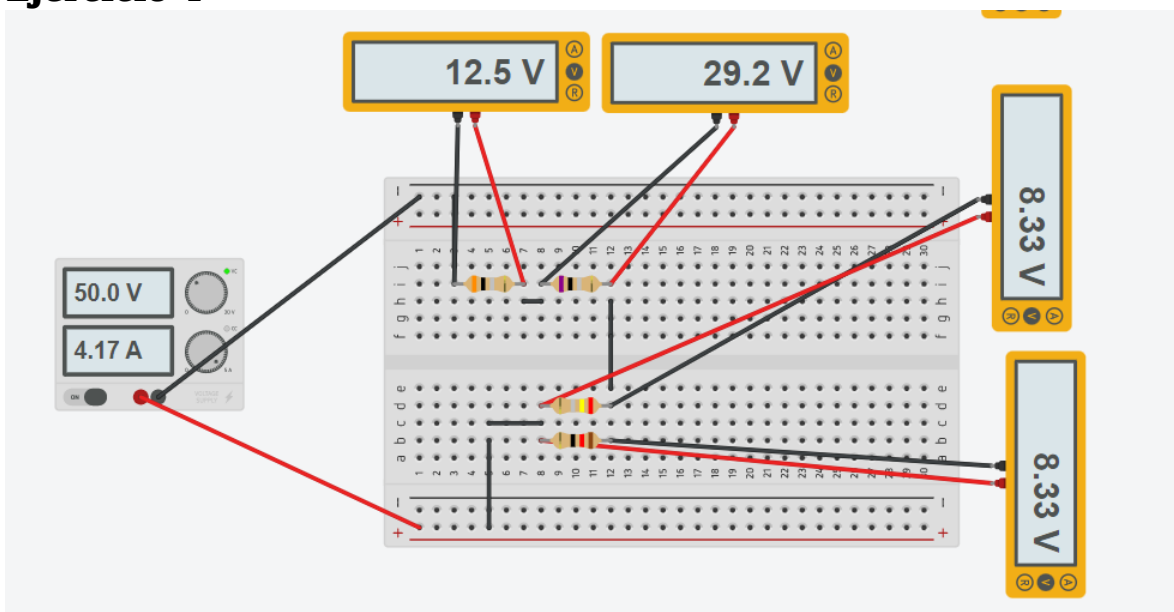
Ejercicio 2



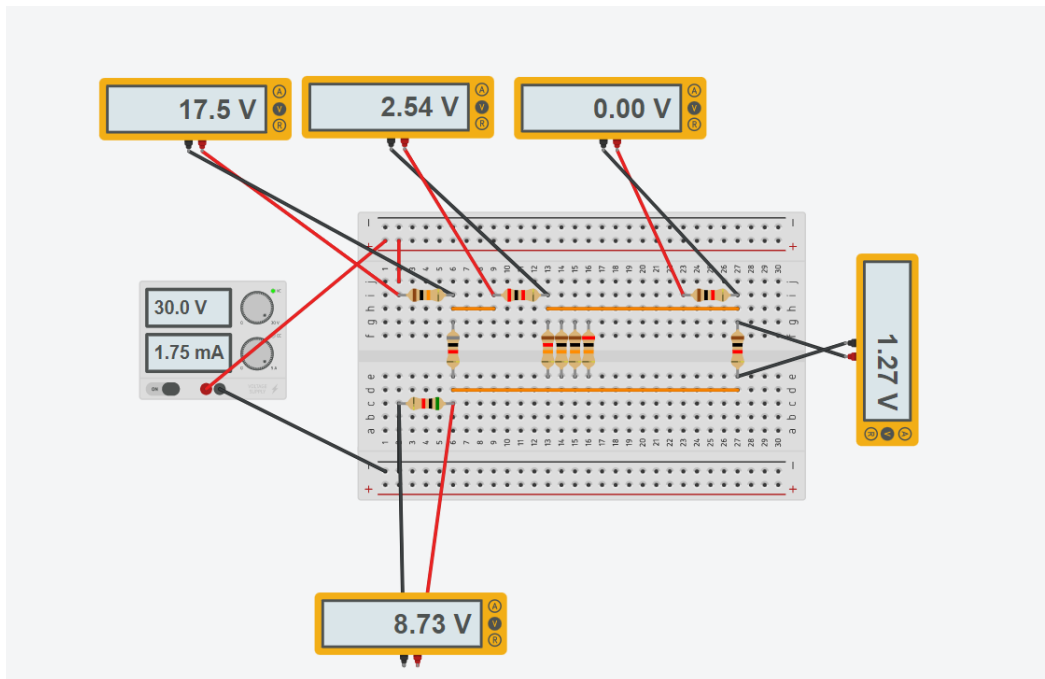
Ejercicio 3



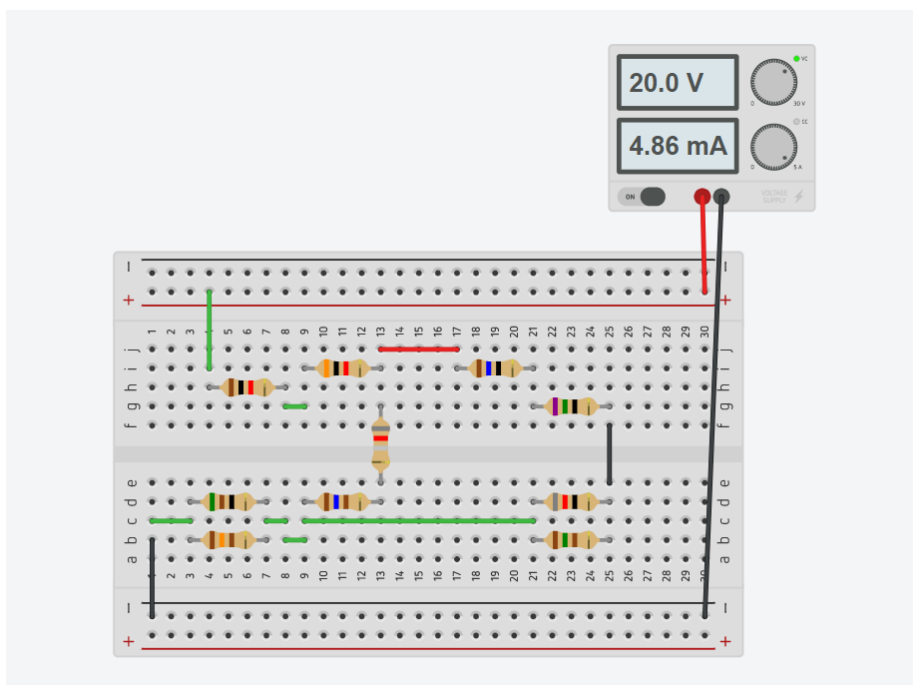
Ejercicio 4



Ejercicio 5



Ejercicio 6



Ejercicio 7

